Programme	:	B.Tech., Integrated	**Semester**	:	Fall Semester 2026-27
Course	:	Computer Networks	**Code**	:	CSE3006
Faculty	:	Dr.S.Periyanayagi	**Slot/ Class No.**	:	B14+B23+D21, BL2026270100337
Tutorial No.	:	2	**Max. Marks**	:	50 Marks
Date of Submission: 19.08.2026					

1. What is the Hamming distance for each of the following codewords? [CO2, KL2] (5 Marks)
   a. d (10000, 00000)
   b. d (10101, 10000)
   c. d (00000, 11111)
   d. d (00000, 00000)
2. Assuming Odd & Even parity, find the parity bit for each of the following data units
   a. 1001011 b. 0001100 c. 1000000 d. 1110111 [CO1 & CO2, KL2] (5 Marks)

3. Using the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $x^3 + x^2 + x$. Find the value of check bits $c_2c_1c_0$ appended at the end of the message by the transmitter using the above CRC scheme and verify at receiver end the code word to get the original data or not. consider the message $m_5m_4m_3m_2m_1m_0 = 110000$ are to be transmitted. [CO2, KL3] (10 Marks)
4. Given the data word of 1000 for a (7,4) hamming code, show the generation of the codeword using even parity. Assume that at the receiver, the b_3 bit in the codeword is received with an error. With suitable illustration, explain the error detection and correction process using the decoder. [CO2, KL3] (10 Marks)
5. a. A sender transmits the following data: 11111011111101111110. Perform **bit stuffing** and write the transmitted bit stream.
   b. Given: Flag (F) = 01111110 , Escape (ESC) = 01111101
   Data (Binary Bytes) to be transmitted

Byte 1	a. Byte 2	b. Byte 3	c. Byte 4	d. Byte 5
01000001	e. 01111110	f. 01000010	g. 01111101	h. 01000011

Perform byte stuffing and write the transmitted frame. [CO2, KL3] (10 Marks)

6. . The sender wants to transmit the following 8-bit data words: [CO2, KL3] (10 Marks)
   D1: 10101010
   D2: 11001100
   D3: 11110000
   Using checksum, calculate:
   a. The checksum.
   b. The transmitted frame.
   c. Verify the checksum at the receiver with error (Assume D2: MSB bit is error and without error.